

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, Berkeley, CA

Aug 2025 - present

- Graduate researcher in the Bay Area Neutron Group
- Researching experimental techniques for indirect neutron capture cross-section measurement of short-lived nuclei

Lawrence Livermore National Laboratory - *Graduate Intern*, Livermore, CA

June 2025 - present

- Developed CRISP (CRITICAL Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP successfully automated the materials compiling and k_{eff} benchmarking of ~3420 ICSBEP critical reactors

Brookhaven National Laboratory - *Nuclear Science Intern*, Upton, NY

2022 - 2024

- Developed BRR (Bayesian Resonance Reclassifier) for heavy-nuclei resonance reclassification on NNDC clusters
- Applied Python machine learning using Scikit-learn and Pytorch to classify resonance spin assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used nuclear data and Random Matrix Theory to develop synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

Truman State University - *Student Astronomy Researcher*, Kirksville, MO

2021 - 2022

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI
- Researched long-exposure luminosity data corruption due to satellite interference